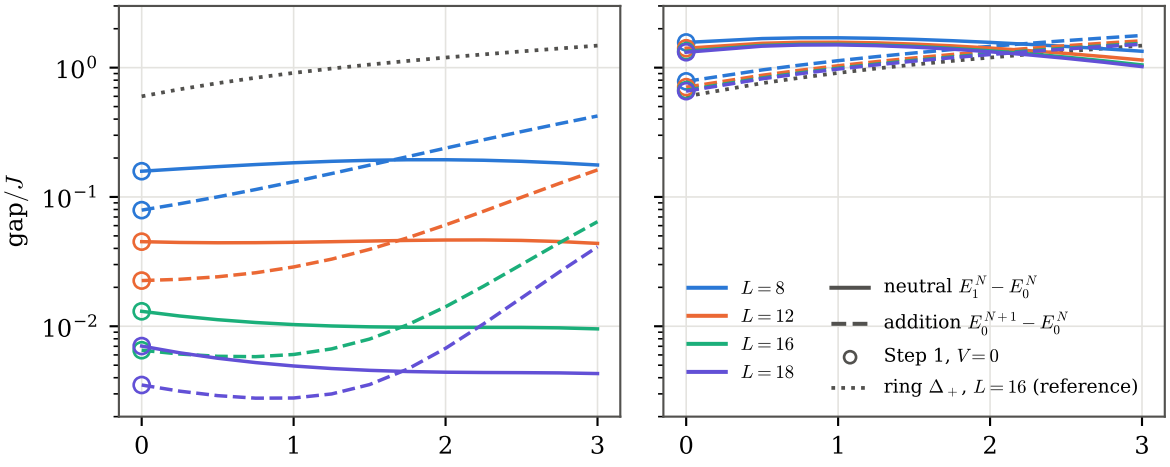
(a) $\delta = -0.30$ (weak end bonds)
(b) $\delta = +0.30$ (strong end bonds)
gap$/J$
10^{0}
10^{-1}
10^{-2}
0
1
2
3
V/J
$L = 8$
$L = 12$
$L = 16$
$L = 18$
neutral $E_1^N - E_0^N$
addition $E_0^{N+1} - E_0^N$
Step 1, $V = 0$
ring Δ_+, $L = 16$ (reference)